

Lecture - 8

Determinants

Let A be an $n \times n$ matrix. Determinant of A is defined as follows

1. $n=1$ Then $A = (a)$; 1×1 matrix. We define
 $\det(A) = a$ $\det \rightarrow$ determinant

2. Suppose that we know how to calculate determinant of any $(n-1) \times (n-1)$ matrix. Let A_{ij} be the matrix obtained from A by deleting the i^{th} row and j^{th} column. Then

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \dots + (-1)^{n-1} a_{1n} \det(A_{1n})$$

Examples

① $A = (2) \Rightarrow \det(A) = 2.$ 1 term

② $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \Rightarrow \det(A) = 2 \det(4) - 3 \det(1)$
 $= 2 \cdot 4 - 3 \cdot 1 = 8 - 3 = 5$

③ $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \boxed{\det(A) = ad - bc}$ 2 terms

(4)

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} \det(A) &= a_{11} \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12} \det \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13} \det \begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \\ &= a_{11} (a_{22} a_{33} - a_{23} a_{32}) - a_{12} (a_{21} a_{33} - a_{23} a_{31}) + a_{13} (a_{21} a_{32} - a_{22} a_{31}) \\ &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} - a_{12} a_{21} a_{33} + a_{12} a_{23} a_{31} + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31} \end{aligned}$$

(6 terms)

first row
second row
third row

Observations

- ① A determinant of an $n \times n$ matrix will have $n(n-1)(n-2) \dots 1$ distinct terms
- ② Each term in the determinant is a product of n entries of A .
- ③ Each term is a product of n entries with exactly one entry coming from each row and each column

product of 3 entries

Proposition 1 $\det(I_n) = 1$. (H.W) Need to use induction (see below)

Proposition 2 Let A be an $n \times n$ matrix. If B is the matrix obtained by replacing the i^{th} row by αR_i then $\det(B) = \alpha \det(A)$

Proof

From observation ③, each term has α appearing exactly once. Therefore, we may simply factor α and obtain $\det(B) = \alpha \det(A)$.

Mathematical Induction

This is a principle that allows us to prove certain statements in mathematics. These statements typically involve a variable which takes positive integer values.

We shall explain this principle through an example.

Example ①

Verify that $1 + 2 + \dots + n = \frac{n(n+1)}{2}$.

Step ①

Verify for $n=1$.

$1 + 2 + \dots + n$ is the same as 1. (LHS)

$$\frac{n(n+1)}{2} = \frac{1 \cdot 2}{2} = 1 \quad (\text{RHS})$$

Since $\text{LHS} = \text{RHS}$, Step ① is Verified!

Step ②

Assume that Step $n-1$ is Verified. That is,

$$1 + 2 + \dots + n-1 = \frac{(n-1)n}{2}.$$

Step ③

Verify step n , which is the same as solving the problem.

We observe that

$$\begin{aligned}
 1+2+\dots+n &= 1+2+\dots+n-1+n \\
 &= \frac{(n-1)n}{2} + n \\
 &= \frac{(n-1)n + 2n}{2} \\
 &= \frac{(n-1+2)n}{2} \\
 &= \frac{(n+1)n}{2}
 \end{aligned}$$

Thus we have verified the statement $1+2+\dots+n = \frac{n(n+1)}{2}$.

Proof of observation 1

Number of terms appearing in $\det(A)$, where A is an $n \times n$ matrix is $1 \cdot 2 \cdot 3 \cdot \dots \cdot n$

Step 1: $n=1$ $\det(A) = a$ (only one term)

Step 2: Assume that for any $n-1 \times n-1$ matrix, number of distinct terms appearing in $\det(A)$ is $1 \cdot 2 \cdot 3 \cdot \dots \cdot n-1$.

Step 3: Since

$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \dots + (-1)^{n-1} a_{1n} \det(A_{1n})$
 " $a_{11}, a_{12}, \dots, a_{1n}$ are distinct " think of it as variables.

and $\det(A_{ij})$ by assumption has $1 \cdot 2 \cdots n-1$ terms,
→ (Step 2)

$\det(A)$ must have

$$\underbrace{1 \cdot 2 \cdots n-1 + 1 \cdot 2 \cdots n-1 + \cdots + 1 \cdot 2 \cdots n-1}_{n \text{ repetitions}} = 1 \cdot 2 \cdots (n-1)n \quad \text{distinct terms}$$

Note that $a_{11}, a_{12}, \dots, a_{1n}$ do not appear in $\det(A_{ij})$ expansion for any j .

Proof of observation 2

Step ①: $n=1$, $A=(a)$, $\det(A)=a$ (nothing to prove)

Step ②: Assume that for any $(n-1) \times (n-1)$ matrix B , $\det(B)$ has terms that are products of entries of B .

Step ③: Since

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \cdots + (-1)^{n-1} a_{1n} \det(A_{1n})$$

and each term of $\det(A_{ij})$, by assumption, is a product of entries of A_{ij} ,
→ (Step 2)

Thus obtain that $a_{ij} \det(A_{ij})$ is a product of entries of A .

Proof of Observation ③

Hints

⊗ Assume that A_{ij} has terms, which are products of n entries from A , each of which comes from exactly one column and exactly one row

For Example $n=2$:

2 entries

$$a_{11}a_{22} - a_{12}a_{21}$$

1st row
1st column

2nd row
2nd column

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$n=3$

$$a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

3 entries

1st row, 2nd column

2nd row 1st column

3rd row 3rd column

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Now, to prove ③, observe that

Ⓐ

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \dots + (-1)^{n-1} a_{1n} \det(A_{1n})$$

Ⓑ Each A_{ij} is an $(n-1) \times (n-1)$ matrix so by assumption, A_{ij} has desired properties. (as written in ⊗).

Since A_{ij} is the matrix obtained by deleting the 1st row and

j^{th} column, the terms of $\det(A_{ij})$ will not have any entry from the 1^{st} row of A and also from the j^{th} column of A .

Now from $\det(A)$ formula, rest of the proof follows!